

Technical Operational Scenario

Autonomous Construction Site Monitoring

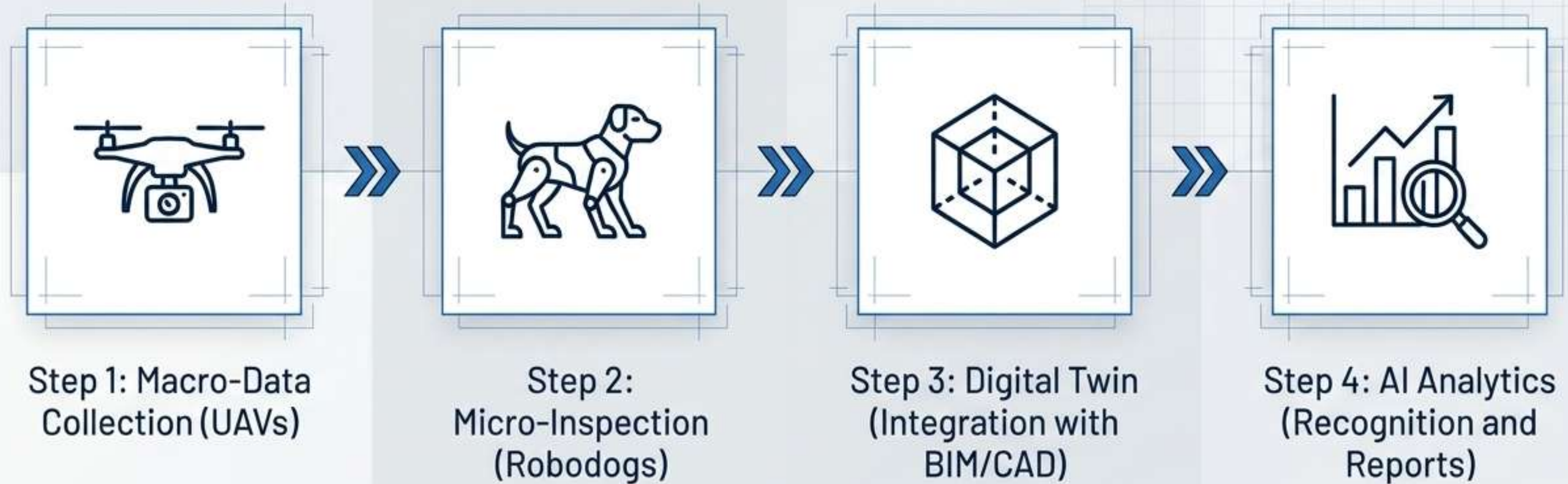
Unified Analytical Platform:
UAVs, Robotics, BIM, and AI.



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Architecture of Autonomous Site Control



Step 1: Aerial Survey Forms a Digital Imprint of the Site

Automatic UAV flights along pre-set routes continuously create an up-to-date copy of the site.



Photogrammetry and
orthomosaics



Building an up-to-date
3D model



Equipment and personnel
control



Real-time cloud data
transfer



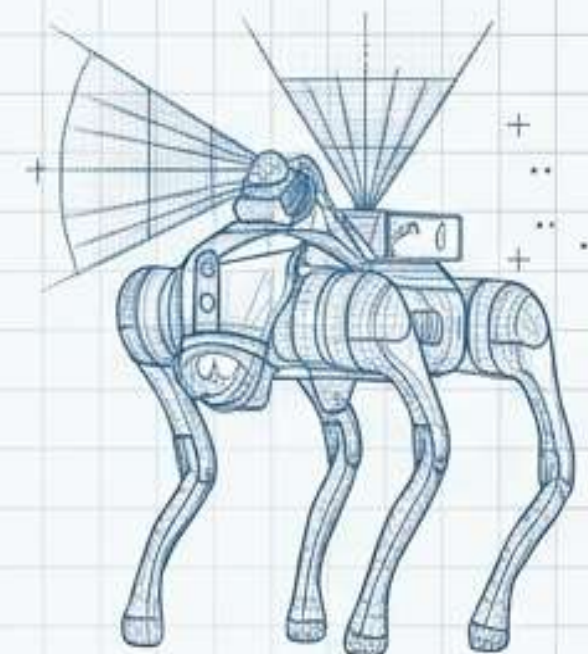
Step 2: Robotic Inspection of Enclosed Zones and Hidden Works

Following the aerial survey, the platform performs deep scanning of interior spaces using LiDAR, HD cameras, and autonomous positioning algorithms.

- ⊕ Quality control of hidden works
- ⊕ Inventory of construction materials in warehouses
- ⊕ Detecting deviations from blueprints
- ⊕ Autonomous indoor navigation



Comprehensive Coverage Achieved Through Macro- and Micro-Scanning Synergy



Parameter	Aerial Scanner (UAV)	Ground Scanner (Robot)
Coverage Zone	External contour and master plan	Internal rooms and floors
Collected Data Type	Orthomosaic, 3D wireframe	Point LiDAR scans, HD photos of nodes
Key Task	Volume and logistics control	Quality control and defect detection

Step 3: Comparing Physical Reality with the Design BIM Model

The obtained actual data is automatically overlaid onto CAD drawings and the work production schedule.

1.

Actual volume of completed work

2.

Amount of used materials

3.

Deviations from design parameters

4.

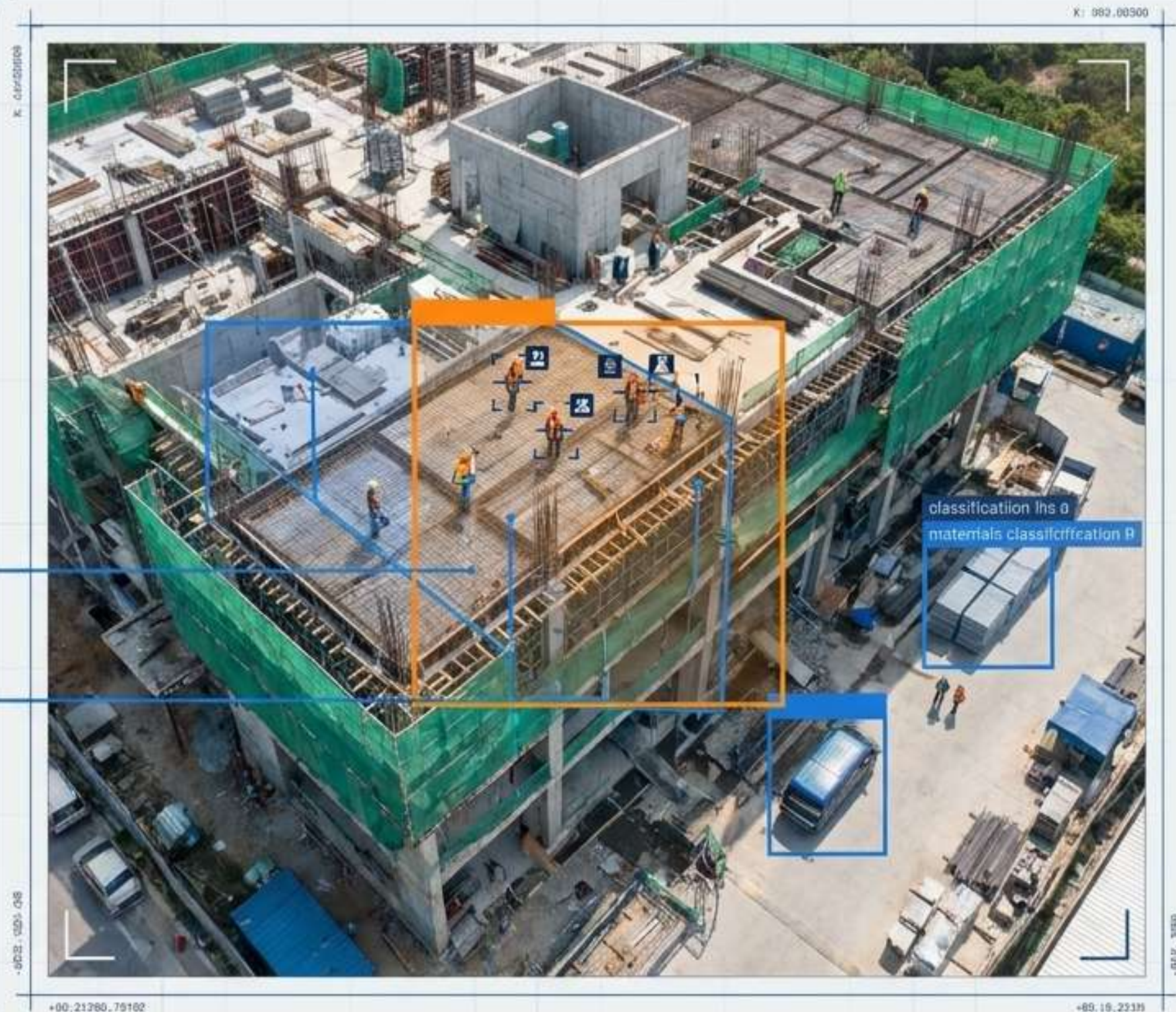
Adherence to the construction schedule



Step 4: Neural Networks Classify Objects and Detect Violations

Artificial intelligence algorithms analyze visual data to automatically generate analytical reports.

- ⊕ Calculating poured concrete volume
- ⊕ Tracking installed elements and mounted structures
- ⊕ Recording construction defects
- ⊕ Identifying safety violations



Operational Summary: Real-Time Data for Decision Making



Metalwork installation: 82%



Facade elements installation: 54%



Engineering networks setup: 41%



Utilization of construction materials: **72%**

 **Calendar schedule: Deviation +5 days**

Automation of Control Drastically Reduces Risks and Costs

-80%

Reduction in site
inspection time

24/7

Daily digital twin
updates without
human presence

100%

Transparency for the
client, contractor,
and investor

The system provides automatic calculation of construction readiness and practically eliminates the human factor when monitoring work execution.

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